WELCOME TO

SET THEORY: UNION, INTERSECTION & VENN DIAGRAMS

WHAT IS A SET?

A set is a collection of things or objects!
Examples: Your favorite fruits, sports you play, or school subjects.
We use curly braces { } and capital letters:
A = {apple, banana, mango}

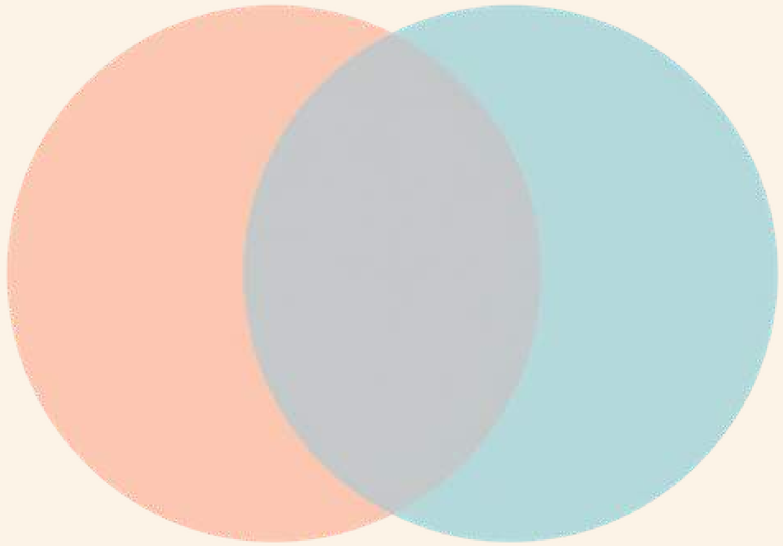
ELEMENTS OF A SET

An element is an item inside a set. For example, if A = {football, cricket, badminton}, then football is an element of set A. We use the symbol "∈" to show this.

football ∈ A means "football is an element of A"

Each item in the curly braces { } is called an element

UNDERSTANDING UNION (A ∪ B)



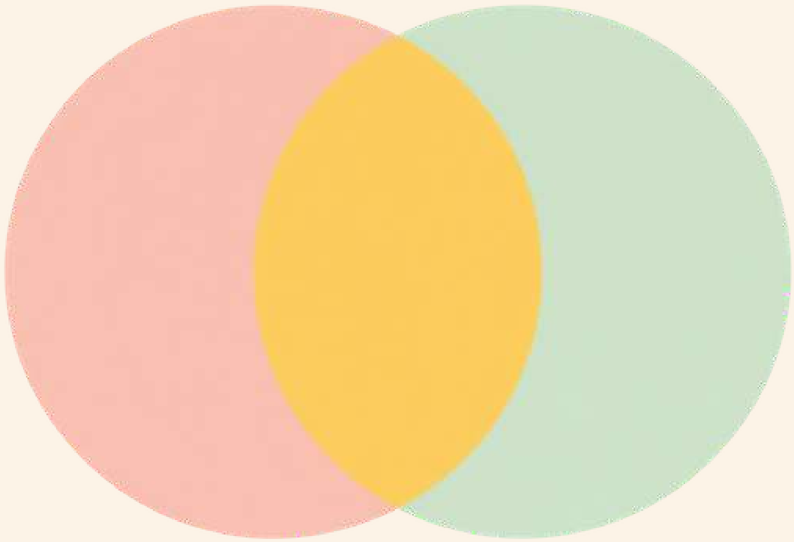
The union of two sets is all the elements that are in A or in B or in both. We write it as A ∪ B.

Example: A = {pizza, burger}, B = {burger, pasta}

A ∪ B = {pizza, burger, pasta}

UNDERSTANDING INTERSECTION (A ∩ B)

The intersection of two sets contains only the elements that are common to both sets. It's like finding what friends share in common!



Example: $A = \{pizza, burger\}$, $B = \{burger, pasta\}$

$A \cap B = \{burger\}$ — only burger is in both sets!

WHAT IS A VENN DIAGRAM?

A Venn diagram is a picture that shows relationships between sets using circles. Each circle represents a set, and when circles overlap, the overlapping area shows elements that belong to both sets. The parts that don't overlap show elements unique to each set.

Reading Venn Diagrams – Step by Step

Step 1: Identify each set's elements

Step 2: Look at overlapping area (intersection)

Step 3: Look at whole shaded area (union)

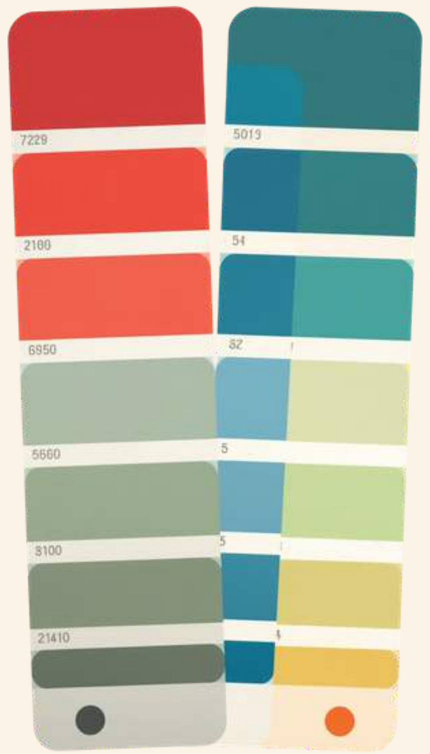
Step 4: Label elements correctly

Step 5: Practice with examples

Follow these five steps to understand any Venn diagram. Practice makes perfect!

Practice Problem 1: Pets & Animals

A = {cat, dog, rabbit}
B = {dog, fish}
Find A ∪ B = ?
(Hint: List ALL elements from both sets)
Find A ∩ B = ?
(Hint: What element is in BOTH sets?)



Practice Problem 2: Colors

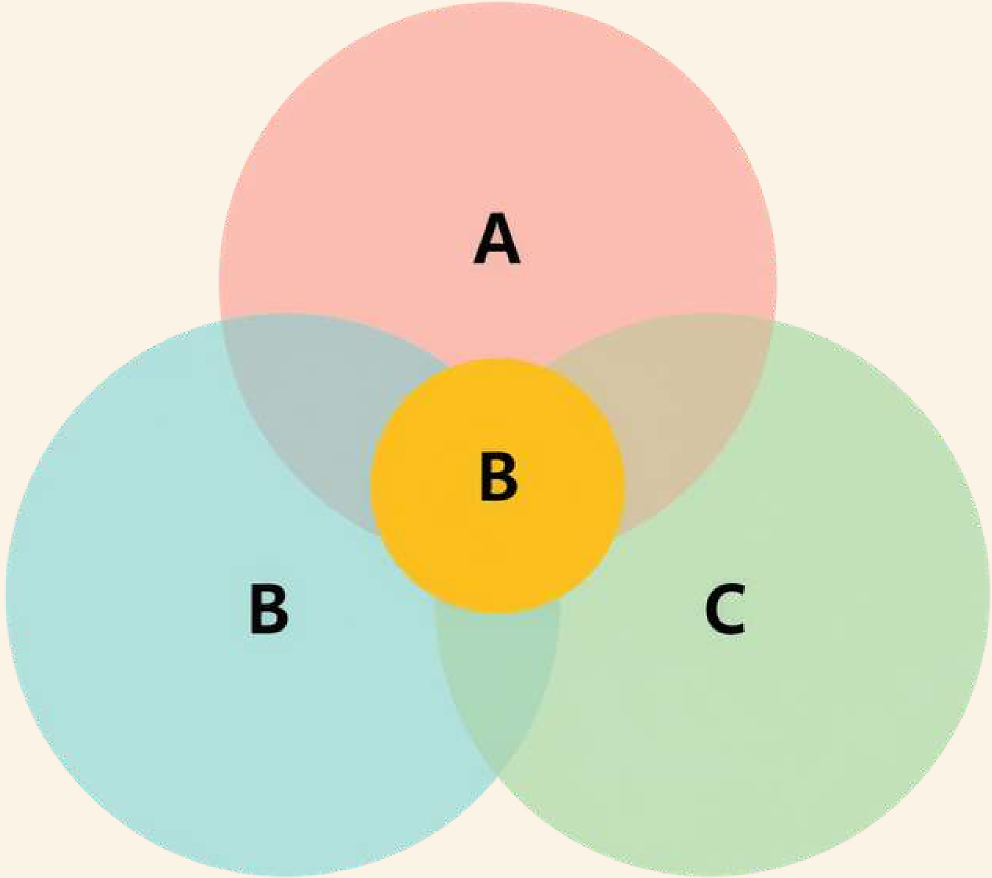
A = {red, blue}
B = {green, yellow}
Find A ∪ B = ?
Find A ∩ B = ?
(Hint: Do these sets share any colors?)

MORE COMPLEX VENN DIAGRAMS – THREE SETS

When we have three sets A, B, and C, we use three overlapping circles.
Each area shows different combinations:
• Only in A, B, or C (outer parts)
• Shared by two sets (where two circles overlap)
• Shared by all three (the center where all meet)
Example: A = {Cricket, Football}, B = {Football, Badminton}, C = {Badminton, Cricket}

UNION & INTERSECTION WITH THREE SETS

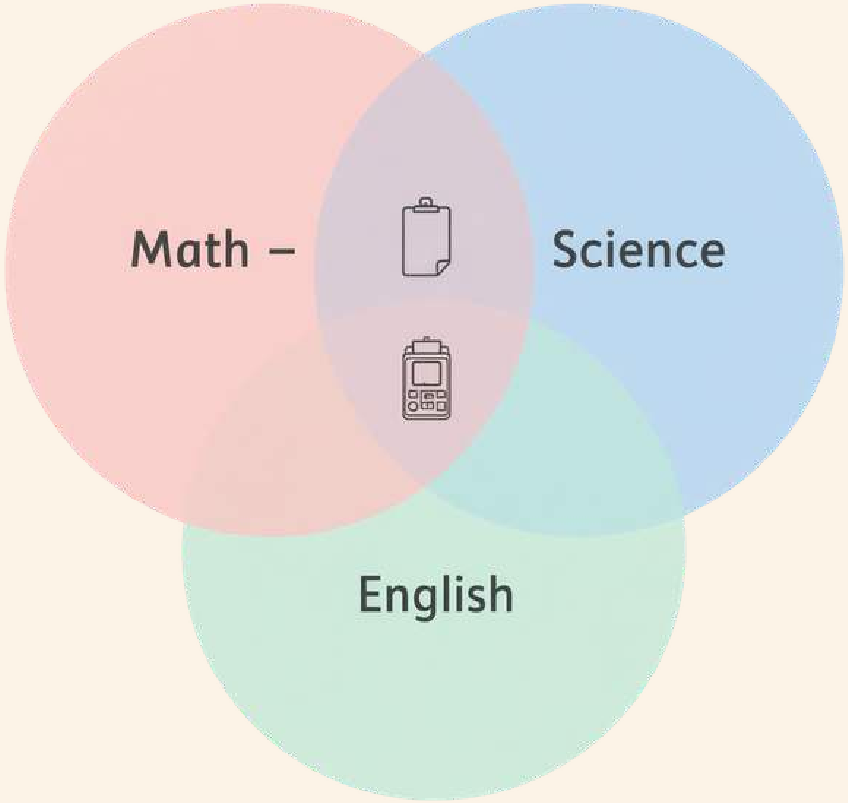
When we have three sets A, B, and C:
• Union (A ∪ B ∪ C) = all elements in any of the three sets
• Intersection (A ∩ B ∩ C) = only elements in ALL three sets



A ∪ B ∪ C = {dancing, singing, drawing, reading, painting}

A ∩ B ∩ C = {drawing} — the hobby everyone shares!

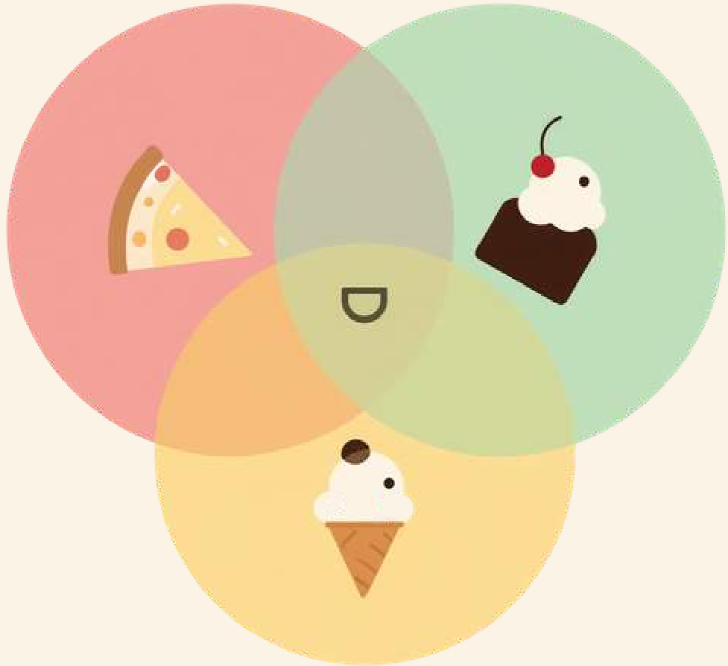
REAL-LIFE EXAMPLE: SCHOOL SUBJECTS



A = Students who like Math
B = Students who like Science
C = Students who like English

Using Venn diagrams, we can find students who like any subject (union) or all subjects (intersection). The overlapping center shows students who love Math, Science, AND English!

FUN WITH FOOD CHOICES



Let's use sets to see what foods our classmates love! Can you find the union and intersection?

P = {pizza lovers}, I = {ice cream lovers}, C = {chocolate lovers}. $P \cap I \cap C$ = friends who love ALL three treats!

Why Learn Set Theory?

Set theory isn't just for math class! We use sets every day to organize, sort, and make decisions. Here are some fun real-world uses:

Sorting sports teams by skill or age	Planning parties and guest lists	Making everyday decisions
Organizing hobbies and activities	Computer science basics for data	Finding friends' favorite things

Summary: What We Learned

A set is a collection of objects, and elements are items inside a set

Union (A ∪ B) means all elements from both sets combined

Intersection (A ∩ B) means only the common elements shared by sets

Venn diagrams help us visualize relationships between 2 or 3 sets

Practice with examples helps us understand set theory better!

Questions & Let's Practice More!

Any questions about sets, union, intersection, or Venn diagrams? Don't be shy — raise your hand! Try more practice problems at home or in class to become a set theory expert.

Teacher Support Available

Your teacher is always here to help! Ask questions during class, visit during break time, or write your doubts in your notebook to discuss later.

SET THEORY

THANK YOU!

Sets are everywhere in life!